

Clinical Motivation

Early atrophy (EA) is an important precursor to advanced atrophic change in age-related macular degeneration (AMD). Clinicians have also observed a vertical hypertransmission pattern in baseline OCT, informally called **barcoding**, which is believed to mark regions at greater risk of subsequent progression.

Barcoding has no standardized quantitative definition. Visual assessment is difficult to reproduce at scale, motivating an interpretable method to localize and measure both barcoding and EA at baseline.

Completed Preprocessing

1. Load a selected B-scan and scanner-provided segmentation metadata from a HEYEX E2E volume.
2. Flatten the scan to BM and retain 150 pixels below it.
3. Standardize the retained region within each scan using a z-score.
4. Apply mild Gaussian denoising (1.0, 0.5) while preserving vertical patterns.

These operations place scans into a comparable anatomical coordinate system while retaining the sub-BM signal used by the detector.

Features and Interpretation

Features are robustly standardized relative to the same scan:

$$z_F(x) = \frac{F(x) - \text{median}(F)}{\text{IQR}(F) + \epsilon}.$$

Intensity. Column median $M(x)$ represents typical hypertransmission, while the 90th percentile $Q(x)$ captures the brightest relevant pixels.

Depth continuity. Horizontal patterns in a local window are correlated between rows separated by the selected lag $d = 4$:

$$C(x) = \frac{1}{D-d} \sum_{y=1}^{D-d} \text{corr}(\mathbf{l}_{y,x}, \mathbf{l}_{y+d,x}).$$

High continuity means that bright and dark locations stay aligned deeper into the scan.

Verticality. Smoothed gradients form a structure tensor $\mathbf{J}$. Its eigenvalues and dominant orientation produce

$$V(x) = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2 + \epsilon} \cos^2 \theta.$$

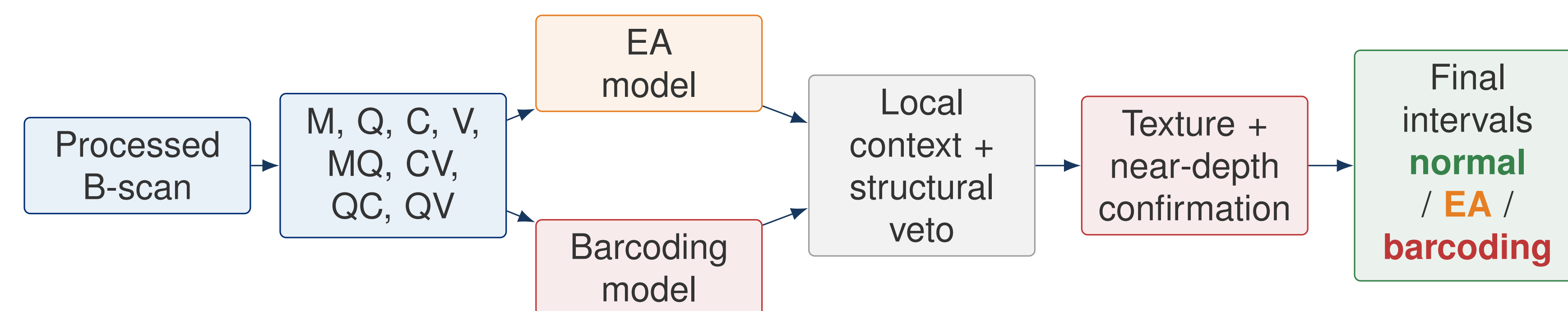
The first term measures directional organization; $\cos^2 \theta$ asks whether that direction agrees with a vertical line. Larger V therefore indicates a stronger vertical structure.

Gabor texture. Quadrature filters measure repeating vertical bright–dark patterns at wavelengths $\lambda = 4, 8, 16$ pixels:

$$G_\lambda(x) = \text{mean}_y \sqrt{(I * g_\lambda^c)^2 + (I * g_\lambda^s)^2}.$$

The calibrated models use $[M, Q, C, V]$ plus interactions after LRT. Positive interaction evidence means two visible properties jointly support a phenotype more strongly than either property alone.

Model Framework



Separate calibrated logistic models score EA and barcoding. Local support, interval probability, feature votes, minimum width, and gap filling remove isolated responses. A structural model and dark-shadow rule reject likely vessels or anatomy. Gabor and near-depth gates provide additional confirmation only for barcoding.

Current Calibrated Specification.

Barcoding

- ▶ At least 55% local support within ± 5 columns
- ▶ Interval mean/peak probability $\geq 0.52/0.68$
- ▶ At least two feature votes
- ▶ Minimum width 8; maximum bridged gap 4
- ▶ Gabor texture and near-depth confirmation

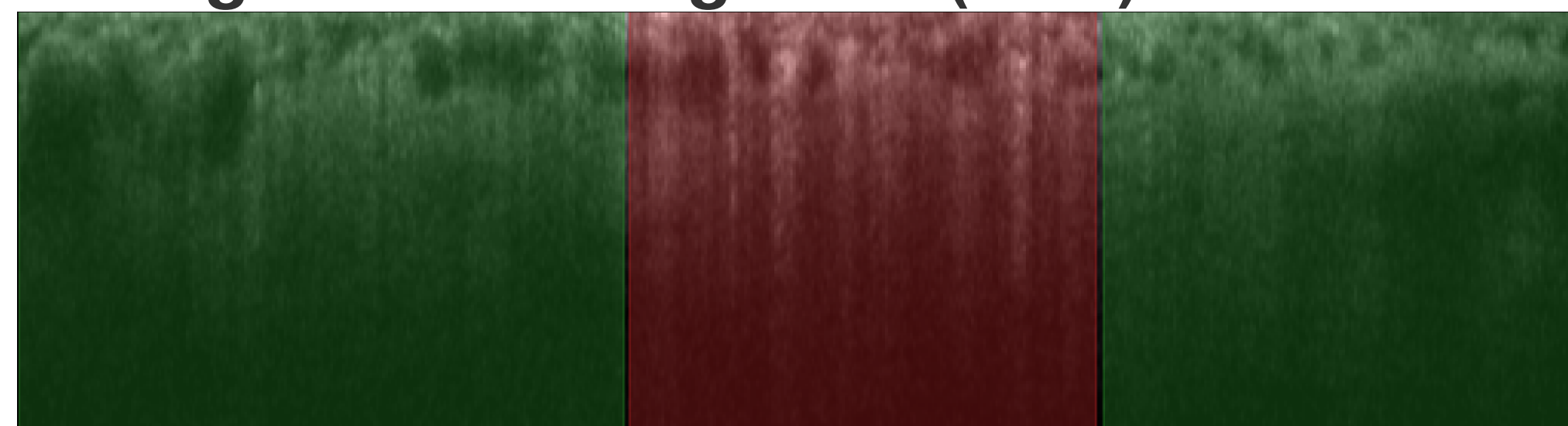
Early Atrophy (EA)

- ▶ At least 60% local support within ± 7 columns
- ▶ Interval mean/peak probability $\geq 0.48/0.62$
- ▶ At least three feature votes
- ▶ Minimum width 12; maximum bridged gap 5
- ▶ No Gabor-texture or depth-band gate

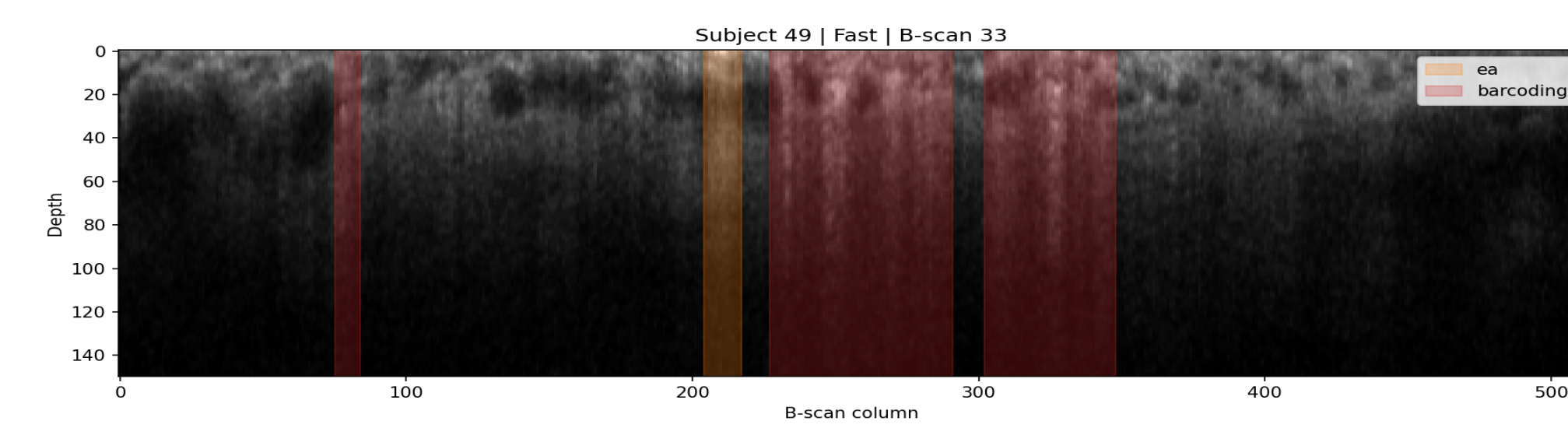
A candidate is retained when it demonstrates sufficiently strong vertical-pattern energy together with increased brightness near Bruch’s membrane. Hypertransmission is evaluated across near (0–20%), middle (20–50%), and deep (50–100%) sub-BM regions to characterize how the signal persists with depth.

Segmentation Results

Strongest Barcoding Case (Fast)



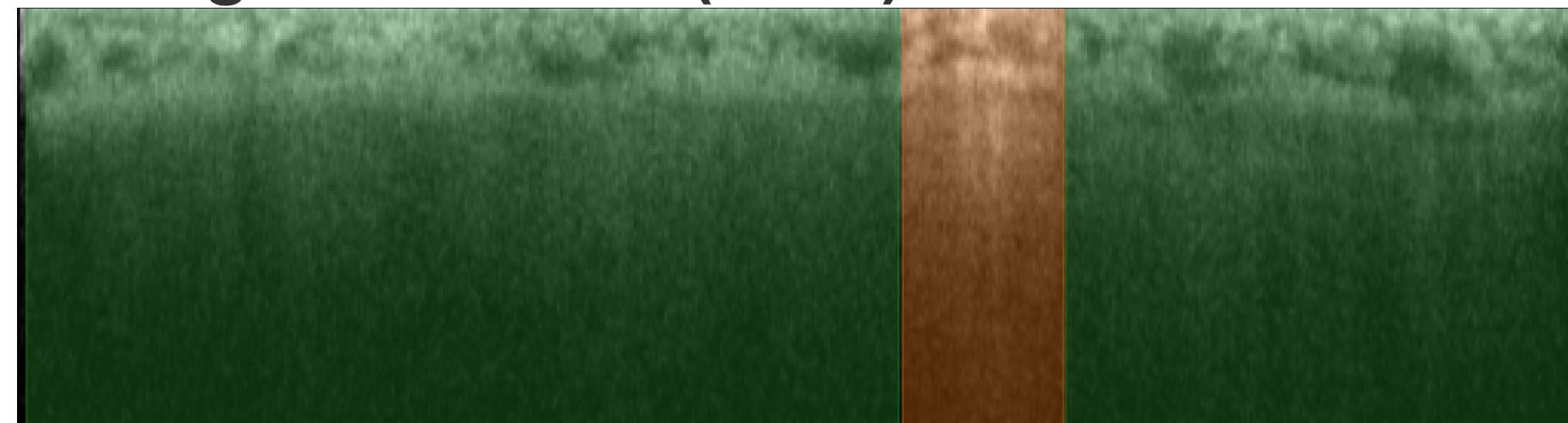
Ground Truth



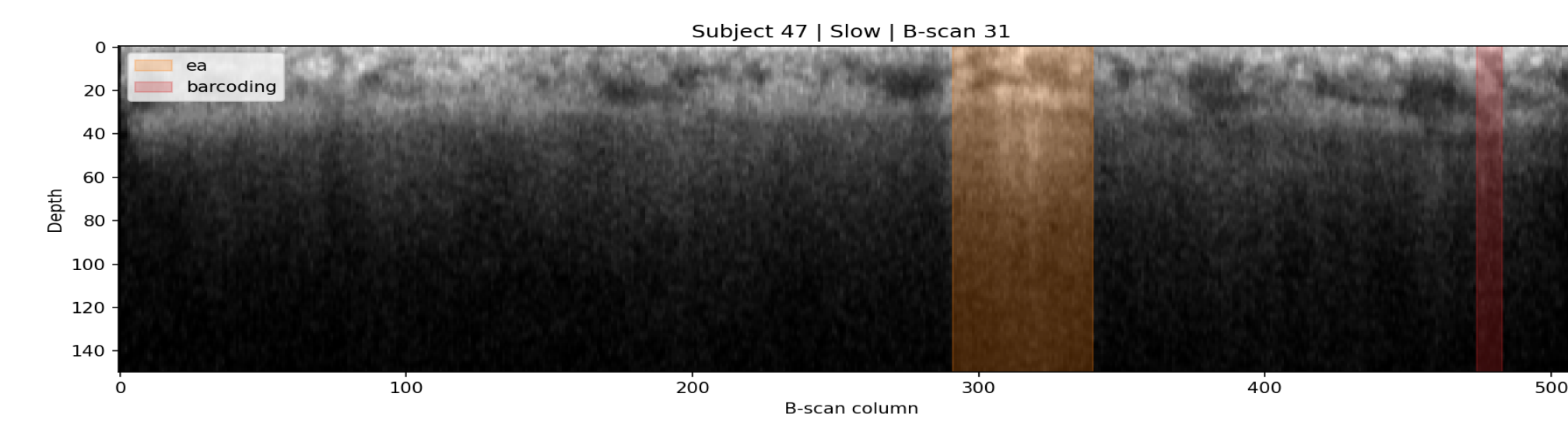
Detector Output

Barcoding precision 0.92, recall 0.71, Dice 0.81.

Strongest EA Case (Slow)



Ground Truth



Detector Output

EA precision 1.00, recall 0.91, Dice 0.95.

These cases show the best saved agreement for each phenotype. The fast case closely captured the major barcoding regions; the slow closely reproduced the single manual EA interval. Small additional phenotype responses remain visible in both automatic outputs, due to small training size.

Ten-Scan Validation and Group Agreement

Five manually selected fast-progressor and slow-progressor B-scans were loaded from the pipeline and compared with manual annotations. Uncertain and vessel/structural columns were excluded from scoring.

Detector Output	Precision	Recall	Dice
Current model: barcoding	0.817	0.520	0.636
Current model: EA	0.462	0.699	0.557

The two rows are outputs from the same pipeline. Precision measures resistance to false alarms, recall measures recovery of manually marked tissue, and Dice summarizes overlap.

Agreement within each five-scan progression group:

Group	Finding	Prec.	Recall	Dice
Fast	Barcoding	0.775	0.545	0.640
Slow	Barcoding	0.303	0.370	0.333
Fast	EA	0.259	0.549	0.352
Slow	EA	0.629	0.760	0.688

Barcoding agreement was stronger in the selected fast scans, whereas EA agreement was stronger in the slow scans. Because these ten selected scans informed detector tuning, all values are development results rather than independent clinical-validation estimates and cannot establish a biological fast-versus-slow difference.

Results and Future Work

Could barcoding be a useful biomarker? In this ten-scan development set, both progression groups had the same number of automatically detected barcoding intervals (18 each), but the fast group had greater total detected barcoding width (451 versus 321 image columns; 40.5% greater mean width per scan). This preliminary pattern suggests that *barcoding extent*, rather than interval count alone, may be worth testing as a progression biomarker. It is not yet evidence of a clinical association because the sample is small, selected, and used to tune the detector. Next steps to take as soon as we get more annotated scans include:

1. **Test correlations with functional measurements.** Evaluate whether barcoding and EA presence, interval count, width, location, texture, and depth persistence are associated with available functional measurements and subsequent progression outcomes.
2. **Develop a hidden Markov model (HMM).** Use spatial context and transitions between neighboring scan columns to improve automated EA and barcoding detection while reducing isolated false-positive regions.

Acknowledgments

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